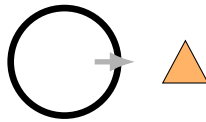


SPROLL 02

What Happened First?



order and the growing history

The Sproll Curriculum · Cycle I: Pops

The Sproll Curriculum

Cycle I — Pops: Things, Differences, and Counting

Sproll 02 of 40 · follows Sproll 01: Something Happened

[Illustration: a bold, solid circle alone on the page]



[Illustration: the same circle, now settled and plain, with a new bold triangle appearing beside it]



[Illustration: the circle and triangle sitting together, both plain and settled, no longer highlighted]



[Illustration: a clean, empty page – a fresh start]

(a clean page) Let's do it again, the other
way.

[Illustration: a bold, solid triangle alone on the page]

 Pop the triangle.

[Illustration: the same triangle, now settled and plain, with a new bold circle appearing beside it]



Then Pop the circle.

[Illustration: the triangle and circle sitting together, both plain and settled – arranged to look exactly like the picture on page 5]



Here is what we have now.

[Illustration: the two final pictures from page 5 and page 9 shown side by side, drawn identically]



Picture One



Picture Two

Are these the same?

[Illustration: the same two pictures, unchanged, now with a small equals sign drawn faintly between them]

Yes.

They look exactly the same.

[Illustration: one of the two identical final pictures, alone, with a small question mark hovering above it]



The picture doesn't say.

[Illustration: two small ledger strips stacked, exactly like the ones from Sproll 01 – the top strip reads circle then triangle in that order, the bottom strip reads triangle then circle in that order]



The history remembers.

Even when the picture doesn't.

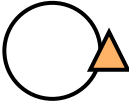
[Illustration: no new image; the two ledger strips from the previous page, shown small and quiet at the bottom of an otherwise empty page]



Same things.

Different history.

[Illustration: the final picture from page 5/9 once more, plain and alone, with the same faint question mark from page 12]

 If the picture can't tell us,

how could we ever know what really happened?

— turn the page for Sproll 03: Play It Again —

A Note for Grown-Ups

This book introduces order without introducing any new primitive. A child already knows what a Pop is from Sproll 01; this book asks what changes when two Pops happen in a different sequence.

The central move is deliberately unsettling in a small, productive way: two different histories, $H_1 = \langle \text{circle, triangle} \rangle$ and $H_2 = \langle \text{triangle, circle} \rangle$, can produce the same final picture. If a child only ever looked at the resulting picture, the two histories would be indistinguishable. But the histories themselves, the ledgers of what was chosen and in what order, are not the same object, even though what they show at the end looks identical.

This book does not yet explain how to recover the order from the picture alone, because in general it cannot always be done from the picture alone. That is a genuine limitation, not a simplification for children, and the curriculum returns to it formally in Sproll 07 once Collapse is available to say precisely what a picture keeps and what it discards.

What this book does establish is the reason a history is worth keeping at all: a picture is already a kind of summary, and a summary can discard exactly the information that mattered. Sproll 03 takes up the natural next question raised here, whether a kept history can be trusted to reconstruct what actually happened.

Sproll 03: Play It Again continues directly from this book's final page.